Permutations

1 Evaluate the following expressions:

a $5!$

b $\dfrac{7!}{5!}$

c ${}_6P_3$

d ${}_5P_5$

e ${}_4P_0$

f ${}_{11}P_1$

g ${}_7P_3$

h ${}_8P_6$

2 How many ways can 9 different items be arranged in a line?

3 Five cards have different letters written on them. The letters are M, L, I, E, S. The cards are shuffled and laid out on a table with the letters face up next to one another. How many possible arrangements are there?

4 How many different ways can the letters of the word 'SHELF' be arranged?

Probabilities with permutations

5 The digits $3, 6, 9, 2, 4$ and 5 are used to make numbers that contain 2 or more digits, but the number cannot contain repeated digits. What proportion of all possible numbers are even?

6 A four digit number is randomly made using each of the digits $1, 5, 4$ and 7 once.

a How many different four digit numbers can be made?

b How many four digit numbers divisible by 5 can be made?

c What is the probability that the number is divisible by 5?

7 The letters of the word ANKLET are randomly arranged in a line.

a What is the probability that the letters L and T will be together?

b What is the probability that there are exactly 2 letters between the letters L and T?

8 The letters of the word CINNAMON are randomly arranged in a line.

a What is the probability that all three N's will be together?

b What is the probability that exactly two N's will be together?

9 The letters of the word SPACE are to be rea[redacted]ed.

a How many different arrangements are possible?

b What is the probability that the letter E will be the first letter?

c What is the probability that the letters are arranged in alphabetical order?

10 8 cards have different letters written on them. The letters are A, R, I, O, S, C, G, U. The cards are shuffled and laid out on a table with the letters face up next to one another.

a How many possible arrangements are there?

b What is the probability that the letters will spell the word GRACIOUS?

11 A random three-letter word is to be formed from the letters A, B, C, D, E and F without repetition.

a What is the probability that the word will begin with a vowel?

b What is the probability that the word will end with a consonant?

c What is the probability that the word will consist of all vowels?

d What is the probability that the word will consist of all consonants?

e What is the probability that the word will consist of a combination of vowels and consonants?

12 A random number is to be formed from the digits 1, 2, 3, 4 and 5 without repetition.

a What is the probability that the number will be $45\,123$?

b What is the probability that the number will have digits that are in ascending order?

c What is the probability that the number will have digits that are in descending order?

d What is the probability that the number will be even?

e What is the probability that the number will be odd?

f What is the probability that the number will be less than $40\,000$?

Applications

13 If there are 5 swimmers in a race, in how many different orders can they finish (assuming there are no ties)?

14 If an mp3 player contains 7 songs, in how many different ways could all 7 songs be played,

15 Christa, James, Beth, Buzz and Eileen are to be placed in a line in random order.

a What is the probability that they will be arranged from tallest to shortest?

b What is the probability that they will be arranged from oldest to youngest?

c What is the probability that they will be arranged in alphabetical order?

d What is the probability that the boys will be next to each other and the girls will all be next to each other?

16 In Arkansas, standard motor vehicle number plates consist of three digits followed by three letters.

a How many different number plates are possible?

b What is the probability that the number plate will be $246\ \mathrm{STZ}$?

c What is the probability of the number plate ending with L or B?

17 To determine the successful applicant for a job, the interviewer assigns a problem to each of the 6 applicants.

a If the problems are assigned at random to each applicant, how many different ways can they be assigned?

b In how many ways can the problems be assigned if Beth is to receive the most difficult one?

c What is the probability that Beth receives the most difficult problem?

18 An mp3 player contains 90 songs, of which one is Silent Night. The songs are to be played in a random order (so that each song can be played any number of times). Find the chance that Silent Night will be:

a The $2\mathrm{nd}$ song played.

b The $7\mathrm{th}$ song played.

c Either the $2\mathrm{nd}$ or $7\mathrm{th}$ song played.

d One of the first 40 songs to be played.

19 For the multiple choice section of an end of year exam, the examiner noted that there are 5 questions whose answer is A, 2 questions whose answer is B, 6 questions whose answer is C and 7 questions whose answer is D. To avoid students guessing an answer based on the pattern of answers, she randomly orders the questions. When she looks at the final paper where the multiple choice questions are randomly ordered, she sees that all the first 5 answers are A, the next 2 answers are B, the next 6 answers are C and the last 7 answers are D. What is the probability of this order being randomly generated?

20 12 delegates from various countries are randomly seated for round table discussions. Two particular delegates have previously had a confrontation and would prefer not to sit next to each other. What is the probability that they are not seated next to each other, nor directly opposite one another?

21 Three friends run into one another at the airport and discover that they will be on the same flight. The plane they will be flying in consists of 41 rows of seats, where each row has three seats on each side of a central aisle. If all passengers have previously been randomly allocated seats, find the probability that:

a All three friends will sit together.

b At most two of them sit together.

22 In a children's memory test, 7 different toys are arranged in a line. The teacher taps 3 of these toys in a random order, and the child has to tap those toys in the same order shown.

a What is the probability that the teacher taps the first 3 toys in order from left to right?

b What is the probability that the teacher taps the last 3 toys in the line?

23 At the opening ceremony of the Olympic Games, a country is represented by 13 athletes who are to walk out onto the stadium in a line.

a What is the probability that the youngest and oldest members of the team will be at each end of the line?

b What is the probability that the 6 female and 7 male athletes will walk out alternately?

24 In a horse race, a trifecta is the name given to predicting the first three horses in their correct order. In a horse race in which there are 10 horses, what is the probability that Michael picks the trifecta?

25 In an oral French exam, each of the 9 students will be asked a different question to respond to. Before the exam, they are given 12 questions from which the 9 questions will be chosen. 5 of these 12 questions are related to the environment. Iain, Roald and Rochelle are particularly strong on talking about the environment. What is the probability that the question each of them will be asked is about the environment?